

Autonomous Instagram Carousel Pipeline

Client Presentation Guide & System Operating Manual | Brand: @DEEPBROTHERSNEWS

PART 1: How to Explain This Project to Your Client

Elevator Pitch: We have built a fully automated, cloud-hosted content publishing engine for @DEEPBROTHERSNEWS. Whenever you have raw photos from an event, product, or news story, you simply drop 2 to 9 photos into Telegram on your phone. Our cloud system on GitHub Actions takes over: it normalizes your images to professional Instagram dimensions, crafts an optimized caption with hashtags using Google Gemini AI, uploads everything securely, and publishes a live Instagram Carousel post—sending you the live post URL in seconds.

Key Value Proposition:

Feature	Client Benefit
■ 100% Mobile Workflow	No need to log into Instagram or transfer files. Upload photos directly via Telegram.
■ 100% Clean Image Quality	Preserves 100% raw image resolution without forced text overlays or quality degradation.
■ Gemini AI Captions	Automatically crafts engaging headlines, body text, and high-reach hashtags for maximum reach.
■■ Zero Server Cost	Runs 100% serverless and free on GitHub Actions cloud infrastructure.
■ Dual Telegram Telemetry	Main bot sends live post link; Admin bot (@Dailykeyupdatebot) receives execution reports.

PART 2: Step-by-Step Operating Manual (Re-Run GitHub Actions)

Step 1: Upload Photos on Telegram (Mobile Phone)

- Open Telegram and chat with your bot.
- Send **/start** (resets old buffer & opens fresh session).
- Attach and send **2 to 9 raw photos**.
- Send **/end** (locks batch for processing).

Step 2: Trigger Workflow on GitHub (Browser)

- Open: github.com/ms3593589-beep/Happydemo/actions/workflows/post.yml
- Click **Run workflow** dropdown menu on the right.
- Leave **■ LIVE PUBLISH TO INSTAGRAM** checked.
- Click the green **Run workflow** button!

Step 3: Receive Output & Verification

- Within 60-90 seconds, your Telegram bot sends the live link: <https://www.instagram.com/p/...>
- Admin bot (@Dailykeyupdatebot) sends execution time & status report (■ SUCCESS).

PART 3: Complete Command Reference

Command / Action	Environment / Target	Description
/start	Telegram Mobile Bot	Resets photo buffer and clears temporary cache.

Send 2-9 Photos	Telegram Mobile Bot	Buffers raw slide images for carousel generation.
/end	Telegram Mobile Bot	Finalizes batch and marks status as active.
python pipeline.py --dry-run	Local PC Terminal	Tests full pipeline offline without publishing.
python pipeline.py	Local PC Terminal	Runs live pipeline execution locally.
python -m unittest discover tests	Local PC Terminal	Executes full test suite (34 unit tests).
git push origin main	Local PC Terminal	Pushes local code changes to Happydemo repo.

PART 4: Configured GitHub Repository Secrets (Happydemo)

Secret Name	Description & Purpose
GEMINI_API_KEY	Google Gemini AI key for caption & hashtag generation.
TELEGRAM_BOT_TOKEN	Main Telegram bot token for receiving photos and post link dispatch.
TELEGRAM_CHAT_ID	Main Telegram Chat ID for receiving live Instagram post URLs.
TELEGRAM_ADMIN_BOT_TOKEN	Dedicated token for @Dailykeyupdatebot admin health reports.
TELEGRAM_ADMIN_CHAT_ID	Admin Chat ID for receiving status & rate limit reports.
CLOUDINARY_URL	Cloudinary cloud media storage for generating public HTTPS image links.
IG_USER_ID	Instagram Professional User ID (17841422784863270).
IG_ACCESS_TOKEN	Long-lived Meta Graph API OAuth access token.